



# Smog and risk of overall and type-specific cardiovascular diseases: A pooled analysis of 53 cohort studies with 21.09 million participants

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## ABSTRACT

The reported associations of smog with the risk of cardiovascular disease (CVD) and CVD subtypes were inconsistent. We systematically searched the Pubmed (Medline) and Embase databases (from the inception to April 25, 2018) to identify the cohort studies investigating the association between smog and CVD and specific types of CVD. We conducted a meta-analysis for different types of air pollutants (PM<sub>2.5</sub>, PM<sub>10</sub>, NO<sub>2</sub>, and O<sub>3</sub>) in smog with the risk of specific types of CVD separately. We summarized the study-specific effect estimates using both the fixed effect model and the random effect model. The meta-analysis included 35 publications with 53 cohort studies. Overall, the associations between per 10 µg/m<sup>3</sup> increase in PM<sub>2.5</sub> exposure and risk of CVD events, stroke events, ischemic heart disease (IHD) events were significant, with relative risks (RRs) of 1.11 (95% confidence interval: 1.07–1.15), 1.12 (95% CI: 1.08–1.16) and 1.14 (95% CI: 1.08–1.21), respectively. PM<sub>2.5</sub>, PM<sub>10</sub>, NO<sub>2</sub>, and O<sub>3</sub> exposure were associated with an increased risk of CVD mortality, with RRs of 1.11 (95% CI: 1.07–1.15), 1.09 (95% CI: 1.02–1.16), 1.23 (95% CI: 1.15–1.31) and 1.03 (95% CI: 1.02–1.05), respectively. Compared with PM<sub>10</sub>, NO<sub>2</sub>, and O<sub>3</sub> exposure, PM<sub>2.5</sub> exposure had a greater risk of stroke incidence and IHD incidence (RR 1.12, 95% CI 1.05–1.19 for stroke incidence; 1.19, 1.09–1.30 for IHD). However, no clear evidence for the associations of PM<sub>10</sub> exposure with risk of CVD incidence, stroke incidence, and IHD incidence was observed. This meta-analysis confirms the evidence that PM<sub>2.5</sub> exposure was significantly associated with increased risk of CVD, stroke, and IHD. PM<sub>2.5</sub>, PM<sub>10</sub>, NO<sub>2</sub>, and O<sub>3</sub> exposure were separately associated with an increased risk of CVD mortality. There was a stronger association between PM<sub>2.5</sub> exposure and the risk of stroke and IHD incidence. It urgently needs well-designed studies to further to elaborate the biological and epidemiological mechanisms that link smog with CVD.

**Main findings:** Compared with PM<sub>10</sub>, NO<sub>2</sub>, and O<sub>3</sub> exposures, PM<sub>2.5</sub> exposure was positively associated with increased risk of stroke and IHD incidence. For air pollutants and CVD events, the association of NO<sub>2</sub> with the risk CVD mortality is more significant.

## 1. Introduction

Smog disperses globally and has been a major public environmental and health crisis worldwide (Landrigan et al., 2018; Lelieveld et al., 2015). Smog is a type of air pollution and attributed to a mixture of smoke and sulfur dioxide. The increasing emissions into atmosphere from industry, fuels, plant powers and vehicles make it easy the formation of particulate matter (PM) and ground level ozone. Those volatile organic compounds and nitrogen oxide by the action of sunlight

tend to contribute to smog (Mukhtar, 2018). The main air pollutants in smog currently recognized linked with health risk are particulate matter (PM) and ozone (O<sub>3</sub>) and nitrogen dioxide (NO<sub>2</sub>) (Fanizza et al., 2018). The effects of smog on health risk are unusually estimated by the effects of the air pollutants in it since smog is complex in the formation, duration, and location (Cai and Wang, 2017; Guan et al., 2017; Mishra, 2017). There have been increasing concerns on the air pollution caused by smog since the Great Smog of 1952. Smog is now referred as the severe air pollution in many countries (Mishra, 2017; Yang et al., 2017;

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Zhang and Samet, 2015). As an important risk factor for health, smog can not only cause an immediate increase in deaths and illness but also contributes to longer-term and more subtle effects (Bates, 2002; Bell and Davis, 2001). Smog has attracted the attention of scholars from various fields, and has been extensively investigated in many aspects, especially in the chemical processes of air pollutants. According to the Global Burden of Diseases Study 2015 (GBD 2015), an estimated 4.2 million global deaths were attributed to exposure to ambient particle mass with the aerodynamic diameter less than  $2.5\ \mu\text{m}$  ( $\text{PM}_{2.5}$ ), which was the fifth-ranking mortality risk factor in 2015 (Cohen et al., 2017b). There is accumulating evidence of the association between exposure to air pollutants and increased risk of chronic obstructive pulmonary disease respiratory infections, impaired lung function, lung cancer and other respiratory diseases (Brugha and Grigg, 2014; Cohen et al., 2017a; Collaborators, 2017). The International Agency for Research on Cancer has classified particulate matter from outdoor air pollution as a group 1 carcinogen that can cause lung cancer (Loomis et al., 2013). Cardiovascular disease (CVD) remains a leading cause of disability, morbidity, and mortality globally (2017). The GBD 2015 reported that in 2015 air pollutants accounted for 19% of all cardiovascular death. In recent years, there are also accumulating investigations of the associations between exposure to smog and risk of cardiovascular disease morbidity as well as mortality. The recent meta-analysis by Adar et al. (2014) provided a systematic evidence of the association between Particulate Matter and Human Health, involving no-accidental mortality, respiratory mortality, respiratory hospital admissions, cardiovascular mortality, cardiovascular hospital admissions, and total mortality.

Although the evidence of the adverse effects of smog on risk of cardiovascular diseases is increasing, the effect estimates of exposure to air pollutants on risk of CVD varied among different studies, and the reported associations of air pollutants with risk of CVD were inconsistent. A recent pooled analysis of 22 cohorts by Beelen et al. (2014) showed the association between long-term exposure to air pollutants and cardiovascular mortality. However, their analyses only investigated the 22 cohorts from Europe, without covering other areas such as Asia and America. The recent meta-analysis by Scheers et al. provided scientific evidence on the association of particulate matters ( $\text{PM}_{2.5}$  and  $\text{PM}_{10}$ ) with the risk of stroke (mortality and incidence), yet it only focused on stroke (Scheers et al., 2015). To provide a more comprehensive, up to date, and detailed assessment of the main air pollutants in smog and the risk of CVD, we conducted a systematic review and meta-analysis of cohort studies investigating the associations of the air pollutants with the risk of CVD and CVD subtypes. We also summarized data on incidence and mortality of CVD and CVD subtypes.

## 2. Method

### 2.1. Search strategy and inclusion criteria

This systematic review and meta-analysis was performed according to the MOOSE guidelines. We systematically searched the Pubmed (Medline) and Embase databases from the inception date to October 23, 2017, and later updated the search to April 25, 2018. The detailed search strategy was shown in Appendix A. We also manually checked for additional studies from the reference list of articles and reviews in our interest. Studies were eligible for inclusion if meeting the following criteria: (1) cohort studies; (2) studies reporting the at least one of the following smog-associated air pollutants exposures:  $\text{PM}_{2.5}$ ,  $\text{PM}_{10}$ ,  $\text{O}_3$ , and  $\text{NO}_2$ ; (3) studies reporting at least one the following outcomes: stroke, ischemic heart disease, heart failure, coronary heart disease,

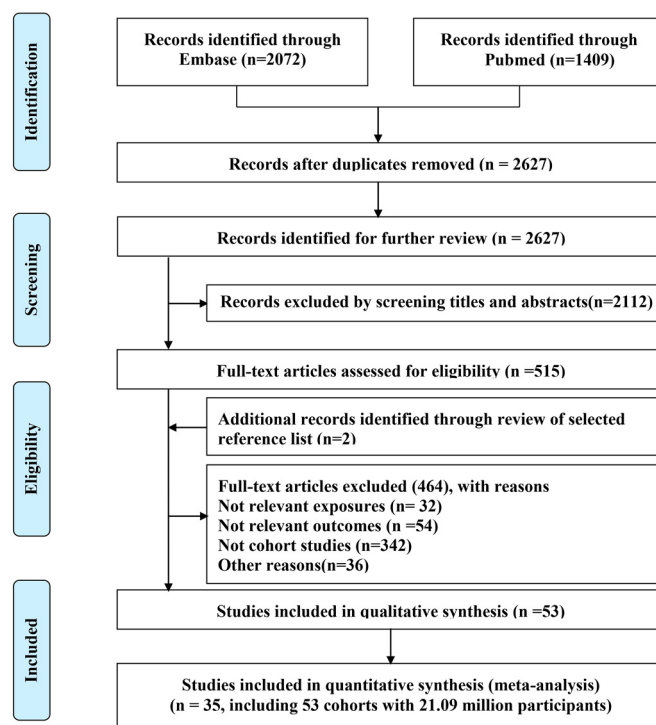


Fig. 1. Flow diagram of study selection.

myocardial infarction, and overall CVD events (mortality, morbidity, or both). (4) studies reporting the effect estimates of the association between the above exposures and outcomes: hazard ratios (HR) from a Cox proportional hazards regression or relative risks (RR) based on incidence/mortality.

### 2.2. Data extraction

Two researchers (HXY and SL) independently extracted the data, the third researcher (YGW) checked the data. Any inconsistencies were resolved by discussion among the three authors. From each eligible article, we extracted information on name of first author, publication year, country, the name of the study, study period, follow-up period, number of participants, sex at baseline, age at baseline, type of exposure, type of amount of air pollutants, type of outcome, number of cases or deaths, relative risks (RR) and 95% confidence intervals (CI), and variables adjusted for in the analysis.

### 2.3. Assessment of risk of bias

We used the Newcastle-Ottawa Scale to assess the risk of bias for eligible studies included in the meta-analysis.

### 2.4. Statistical analysis

We conducted a meta-analysis for different types of air pollutants with the risk of CVD and CVD subtypes separately. We summarized the study-specific effect estimates using both the fixed effects model and the random effects model by Dersimonian and Laird (1986). Hazard ratios, incidence ratios, and relative risks reported in studies were considered as relative risks (RR) in our study. For each study, the RR of the most adjusted estimate and its corresponding 95% confidence interval from each eligible study were selected for the meta-analysis. For

**Table 1**  
Characteristics of included cohort studies in the meta-analysis.

| Author (publication year) | Locations, study name                                              | Study period                        | N (male %)       | Air pollutants                                                          | End point: CVD types (number of cases)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------------|--------------------------------------------------------------------|-------------------------------------|------------------|-------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Atkinson et al. (2013)    | UK, Clinical Practice Research Datalink                            | 2003–2007                           | 831,788(48.3%)   | PM <sub>10</sub> , NO <sub>2</sub> , O <sub>3</sub>                     | Incidence: Stroke(13,102)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Badaloni et al. (2017)    | Italy, the RoLS                                                    | 2011–2016                           | 1,249,108(45%)   | PM <sub>2.5</sub> , PM <sub>10</sub>                                    | Mortality: CVD(59,434), IHD(22,234)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Beelen et al. (2009)      | The Netherlands, the NLCS                                          | 1987–1996                           | 117,528(48.3%)   | PM <sub>2.5</sub> , PM <sub>10</sub>                                    | Mortality: CVD (6137), IHD (3521), CBVD (1175)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Beelen et al. (2014)      | Europe, 22 European Cohorts <sup>b</sup>                           | 1991–2011                           | 367,383(37%)     | PM <sub>2.5</sub> , PM <sub>10</sub> , NO <sub>2</sub> , O <sub>3</sub> | Mortality: CVD(9994), IHD(4992), CBVD (2484)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Benayeb et al. (2015)     | France, the Gazel cohort                                           | 1989–2013                           | 20,327(72.8%)    | PM <sub>2.5</sub> , PM <sub>10</sub> , O <sub>3</sub>                   | Mortality: CVD (165)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Cai et al. (2018)         | Norway, HUNT<br>UK, EPIC-Oxford<br>UK, Biobank                     | 1995–2013<br>1993–2008<br>2006–2011 | 355,732(42%)     | PM <sub>10</sub> , NO <sub>2</sub>                                      | Incidence: CVD (21,081), IHD(3515), Stroke(1230)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Cesaroni et al. (2013)    | Italy, The RoLS                                                    | 2001–2010                           | 1,265,058(NA)    | PM <sub>2.5</sub> , NO <sub>2</sub>                                     | Mortality: CVD(60,318), IHD(22,562), stroke(13,576)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Dan et al. (2015)         | Canada, the CanCHEC                                                | 1991–2006                           | 2,521,525(49.8%) | PM <sub>2.5</sub> , NO <sub>2</sub> , O <sub>3</sub>                    | Mortality: CVD (98,970), IHD(63,050), CBVD(19,275)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Fischer et al. (2015)     | The Netherlands, the DUELS                                         | 2004–2011                           | 7,218,363(47.7%) | PM <sub>10</sub> , NO <sub>2</sub>                                      | Mortality: CVD(209,940)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Hart et al. (2015)        | US, the NHS                                                        | 1989–2006                           | 114,537(0%)      | PM <sub>2.5</sub> , PM <sub>10</sub>                                    | Incidence: CVD(6767), Stroke(3295)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Heinrich et al. (2013)    | UK, the women's cohort in North Rhine-Westphalia                   | 1985–2008                           | 4752(0%)         | PM <sub>10</sub> , NO <sub>2</sub>                                      | Mortality: CVD(268)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Huss et al. (2010)        | Switzerland, The Swiss National Cohort                             | 2000–2005                           | 4,580,311(47%)   | PM <sub>10</sub>                                                        | Mortality: , CVD(177,836), Stroke(25,231)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Katsoulis et al. (2014)   | Greek, EPIC- Greek                                                 | 1997–2011                           | 2752(45.3%)      | PM <sub>10</sub> , NO <sub>2</sub>                                      | Incidence: CVD(212), Stroke(60), IHD(147 )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Kim et al. (2017)         | Korean, the NHS-NSC                                                | 2007–2013                           | 136,094(49.1%)   | PM <sub>2.5</sub> , PM <sub>10</sub> , NO <sub>2</sub> , O <sub>3</sub> | Mortality, CVD(1856), Stroke (934)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Laden et al. (2006)       | US, the Harvard Six Cities Study                                   | 1974–1998                           | 8096(45%)        | PM <sub>2.5</sub>                                                       | Mortality: CVD(1196)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Lin et al. (2017)         | China, Ghana, India, Mexico, Russia, and South Africa, US, the CTS | 2007–2010                           | 45,625(43.2%)    | PM <sub>2.5</sub>                                                       | Incidence: Stroke(1239)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Lipsett et al. (2011)     |                                                                    | 1996–2005                           | 60,204(0%)       | PM <sub>2.5</sub> , PM <sub>10</sub> , NO <sub>2</sub> , O <sub>3</sub> | Mortality: CVD (1630, 1863, 2919 and 408 for PM <sub>2.5</sub> , PM <sub>10</sub> , O <sub>3</sub> and NO <sub>2</sub> , respectively), IHD (773, 843, 1358 and 193 for PM <sub>2.5</sub> , PM <sub>10</sub> , O <sub>3</sub> and NO <sub>2</sub> , respectively), CBVD (382,486,728 and 104 for PM <sub>2.5</sub> , PM <sub>10</sub> , O <sub>3</sub> and NO <sub>2</sub> ,respectively);<br>Incidence : Stroke(969,1179,310 and 254 for PM <sub>2.5</sub> , PM <sub>10</sub> , O <sub>3</sub> and NO <sub>2</sub> , respectively)<br>Incidence: Ischemic stroke(376) |
| Mechtoui et al. (2012)    | France, AVC69 study                                                | 2006–2007                           | 1306(46.3%)      | PM <sub>2.5</sub> , PM <sub>10</sub> , NO <sub>2</sub> , O <sub>3</sub> | Incidence: CVD(1816), Stroke(554)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Miller et al. (2007)      | US, the WHI                                                        | 1994–2000                           | 58,610(0%)       | PM <sub>2.5</sub>                                                       | Mortality: CVD(261), CBVD (122)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Nishiaki et al. (2013)    | Japan, the JPHC                                                    | 1990–2008                           | 62,142(48.7%)    | PM <sub>10</sub>                                                        | Incidence: CVD(2720), IHD(618), Stroke(2181)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Pinault et al. (2016)     | Canada, the community health survey cohort                         | 1998–2000                           | 299,500(46.01%)  | PM <sub>2.5</sub>                                                       | Mortality: CVD(8600), IHD(4700), CBVD(1500)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Qiu et al. (2017)         | China, the Elderly Health Services cohort                          | 1998–2012                           | 61,447(34.1%)    | PM <sub>2.5</sub>                                                       | Incidence: Stroke(6733)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Stafoggia et al. (2014)   | Europe, the ESCAPE project                                         | 1992–2010                           | 99,446           | PM <sub>2.5</sub> , PM <sub>10</sub> , NO <sub>2</sub>                  | Incidence: Stroke(3086)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Stockfelt et al. (2017)   | Sweden, PPS                                                        | 1970–2011                           | 58500(100%)      | PM <sub>2.5</sub> , PM <sub>10</sub>                                    | Incidence: IHD(1826), Stroke(1139)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Thurston et al. (2016)    | Sweden, the GOT-MONIC                                              | 1985–2011                           | 45000(48%)       | PM <sub>2.5</sub> , PM <sub>10</sub>                                    | Incidence: IHD(400), Stroke(252)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| To et al. (2015)          | US, the NIH-AARP Study                                             | 2000–2009                           | 517,041(58.68%)  | PM <sub>2.5</sub>                                                       | Mortality: CVD(26,009)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Tseng et al. (2015)       | Canada, the CNBSS                                                  | 1980–2013                           | 29,549(0%)       | PM <sub>2.5</sub>                                                       | Incidence: IHD(121,069), Stroke(5993)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Turner et al. (2016)      | China                                                              | 1989–2008                           | 43,227(56.91%)   | PM <sub>2.5</sub>                                                       | Mortality: CVD(280), IHD(139), CBVD(141)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Ueda et al. (2012)        | US, the CPS-II                                                     | 1982–2004                           | 669,046(43.8%)   | PM <sub>2.5</sub> , NO <sub>2</sub> , O <sub>3</sub>                    | Mortality: CVD(84,132), IHD(45,644), CBVD(17,085)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                           | Japan, NIPPON DATA80                                               | 1985–2004                           | 7250(43.3%)      | PM <sub>10</sub>                                                        | Mortality: CVD(574), Stroke(250)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

(continued on next page)

Table 1 (continued)

| Author (publication year) | Locations, study name                                  | Study period | N (male %)     | Air pollutants                     | End point: CVD types (number of cases)      |
|---------------------------|--------------------------------------------------------|--------------|----------------|------------------------------------|---------------------------------------------|
| Villeneuve et al. (2015)  | Canada, the CNBSS                                      | 1980–2005    | 89,248(0%)     | PM <sub>2.5</sub>                  | Mortality: CVD(1845), IHD(903), Stroke(434) |
| Weichenthal et al. (2014) | USA, the AHS                                           | 1993–2011    | 83,378(62%)    | PM <sub>2.5</sub>                  | Mortality: CVD(1575)                        |
| Wong et al. (2015)        | China                                                  | 1998–2011    | 59,362(34.2%)  | PM <sub>2.5</sub>                  | Mortality: CVD(4656), IHD(1810), CBVD(1621) |
| Zhang et al. (2011)       | China                                                  | 1998–2009    | 12,584(48.5%)  | PM <sub>10</sub> , NO <sub>2</sub> | Mortality: CVD(138), CBVD(118)              |
| Zhang et al. (2014)       | China                                                  | 1998–2009    | 39,054(49.83%) | PM <sub>10</sub>                   | Mortality: CVD(677), IHD(162), CBVD(295)    |
| Zhou et al. (2014)        | China, nationally representative cohort of Chinese men | 1990–2006    | 71,431(100%)   | PM <sub>10</sub>                   | Mortality: CVD (3869)                       |

Abbreviations: CVD, cardiovascular disease; CBVD, cerebrovascular disease; IHD, ischemic heart disease; PM<sub>2.5</sub>, particle with aerodynamic diameter 2.5 ≤ 10 μm; PM<sub>10</sub>, particle with aerodynamic diameter ≤ 10 μm; NO<sub>2</sub>, nitrogen dioxide; O<sub>3</sub>, ozone.

<sup>a</sup> N: number of participants.

<sup>b</sup> 22 cohorts: Sweden, EPIC-Umeå; Finland, FinRISK; Norway, HUBRO; Sweden, SNAC-K; Sweden, Salt/twin gene; Sweden, 60-y/IMPROVE; Sweden, SDPP; Denmark, DCH; The Netherlands, EPIC-MORGEN; The Netherlands, EPIC-PROSPECT; Germany, SALIA; United Kingdom, EPIC-Oxford; Germany, KORA; Austria, VHM&PP; Switzerland, SAPALDIA; France, E3N; Italy, EPIC-Varese; Italy, EPIC-Turin; Italy, SDIRIA-Turin; Italy, SDIRIA-Rome; Spain, EPIC-San Sebastian; Greece, EPIC-Athens.

studies reporting data separately by cohorts, we treated them as different reports when included in the meta-analysis. The Q and I<sup>2</sup> statistics were used to assess heterogeneity between studies. The Egger's test was used to evaluate publication bias. Subgroup analyses were conducted by characteristics such as geographical region, smoking status, and sex if possible.

The meta-analyses were conducted using Stata version 14.0 software (StataCorp, TX).

3. Results

3.1. Included studies

Flow diagram of study selection was presented in Fig. 1. From identified 2627 records, 854 records were excluded because of duplication. Further screening of titles and abstracts excluded 2112. Then 515 records underwent full-text review, 464 were excluded because 32 were irrelevant exposures, 54 were irrelevant outcomes, 342 were not cohort studies, and 36 were other reasons. Totally, 53 articles were included in qualitative synthesis. Lastly, 35 eligible articles in quantitative synthesis including 53 cohorts with 21.09 million participants were included in this systematic review and meta-analyses. Of these, 25 studies were related to PM<sub>2.5</sub>, 20 studies related to PM<sub>10</sub>, 14 studies related to NO<sub>2</sub>, and 8 studies related to O<sub>3</sub>. The description and detailed characteristics of the included studies were summarized in Table B1, Appendix 2. The eligible articles were published from 2004 to 2018. Of the 53 eligible articles, 17 were based on cohorts from America, 6 from Canada, 18 from Europe, 8 from China, 2 from Japan, 1 from Korea, 1 from multiple areas (China, Ghana, India, Mexico, Russia, and South Africa). Because of the limited data, here, we reported the main results of random effects meta-analyses of 53 cohort studies from 35 eligible articles investigating the associations of air pollutants(PM<sub>2.5</sub>, PM<sub>10</sub>, NO<sub>2</sub>, or O<sub>3</sub>) with the risk of CVD (total CVD events, CVD incidence, CVD mortality, total stroke events, stroke incidence, stroke mortality, total IHD events , IHD incidence, or IHD mortality).

3.2. Summary of cohort studies assessed

All included cohort studies focused on long-term effects of the air pollutants on CVD or CVD subtypes. A summary of all cohorts assessed was described in Table 1. The list of variables included in the "most adjusted" analysis per eligible study was shown in Table B2, Appendix 2. For study quality assessment, most of the included studies had a NOS score of 7 (see Table B2, Appendix 2). The classification for CVD events per included study was shown in Table B2 in Appendix 2. In this meta-analysis, the exposure doses used of PM<sub>2.5</sub>, PM<sub>10</sub>, and NO<sub>2</sub> were per 10 μg/m<sup>3</sup> increase, and the exposure dose used of O<sub>3</sub> was per 9.5–18.4 ppb increase. The association of the air pollutants with the risk of total CVD events, stroke events, and IHD events was investigated in 29, 28, and 17 studies, respectively. The included studies in our meta-analysis covered 21.09 million participants. Here, we reported results of the meta-analyses by the random effect model, considering the heterogeneity.

3.3. Air pollutants and CVD

Per 10 μg/m<sup>3</sup> increase in PM<sub>2.5</sub> was positively associated with increased risk of total CVD events (RR 1.12, 95% CI 1.05–1.19), CVD incidence (RR 1.12, 95% CI 1.05–1.19), and CVD mortality (RR 1.11,95% CI 1.08–1.14), respectively (Fig. 2). There was a weak association of per 10 μg/m<sup>3</sup> increase in PM<sub>10</sub> with the risk of total CVD events (RR: 1.06, 95% CI: 1.00–1.12) (Fig C2, Appendix C). The association of per 10 μg/m<sup>3</sup> increase in PM<sub>10</sub> with CVD mortality is significant, with an RR of 1.09 (95% CI: 1.02–1.16). The RR per 10 μg/m<sup>3</sup> increase in PM<sub>10</sub> was 0.97 (95% CI: 0.65–1.3) for CVD incidence. The associations of per 10 μg/m<sup>3</sup> increase in NO<sub>2</sub> with the risks of total CVD

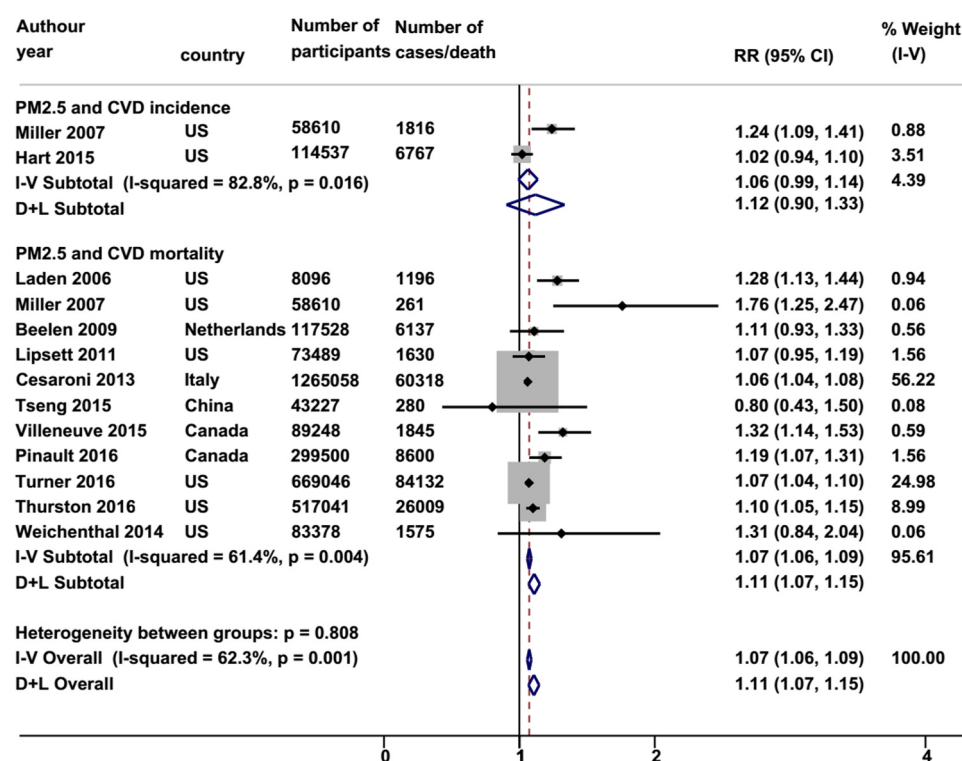


Fig. 2. Relative risk of CVD events (incidence and mortality) per 10  $\mu\text{g}/\text{m}^3$  increase in  $\text{PM}_{2.5}$ . Overall relative risk calculated with both fixed and random effects models.

events and CVD mortality were significant, with RRs of 1.36 (95% CI: 1.09–1.64) and 1.46 (95% CI: 1.13–1.79), respectively (Fig C3, Appendix C). There was a weak association between per 9.5 and 11.0 ppb increase in  $\text{O}_3$  exposure and risk of CVD mortality, with an RR of 1.03 (95% CI: 1.02–1.05) (Fig C3, Appendix C). However, the association between per 9.5 and 18.4 ppb increase in  $\text{O}_3$  exposure and the risk of total CVD events was not significant (RR: 0.93, 95% CI: 0.81–1.04).

### 3.4. Air pollutants and stroke

Per 10  $\mu\text{g}/\text{m}^3$  increase in  $\text{PM}_{2.5}$  was significantly associated with risk of stroke events, with an RR of 1.12 (95% CI: 1.02–1.16). It was also positively associated with stroke incidence and stroke mortality. The RR per 10  $\mu\text{g}/\text{m}^3$  increase in  $\text{PM}_{2.5}$  for stroke incidence was 1.12 (95% CI: 1.02–1.09), and stroke mortality 1.11 (95% CI: 1.07–1.14) (Fig. 3). Our meta-analysis showed there was no significant association between per 10  $\mu\text{g}/\text{m}^3$  increase in  $\text{PM}_{10}$  and the risk of stroke events (RR 1.04, 95% CI: 0.90–1.17), stroke incidence (RR 0.97, 95% CI: 0.83–1.10), and stroke mortality (RR 1.0, 95% CI: 0.86–1.29) (Fig C4, Appendix C). The association between  $\text{NO}_2$  exposure and the risk of stroke disappeared, with an RR of 1.01 (95% CI: 0.98–1.03) for per 10  $\mu\text{g}/\text{m}^3$  increase in  $\text{NO}_2$ . There was also no clear association for stroke incidence and stroke mortality, with RR of 1.02 (95% CI: 0.94–1.10) and 1.08 (95% CI: 0.86–1.29) per 10  $\mu\text{g}/\text{m}^3$  increase in  $\text{NO}_2$ , and stroke mortality 1.11 (95% CI: 1.07–1.14) (Fig C5, Appendix C). We found there was no evidence for the association between per 9.5 and 18.4 ppb increase in  $\text{O}_3$  exposure and the risk of stroke events, with an RR of 0.93 (95% CI: 0.76–1.10). The RR per 9.5–18.4 ppb increase in  $\text{O}_3$  exposure was 0.81 (95% CI: 0.40–1.22) for stroke incidence. There was a weak association between  $\text{O}_3$  exposure and stroke mortality (RR: 1.02, 95% CI: 1.00–1.05) (Fig C6, Appendix C).

### 3.5. Air pollutants and IHD

There was a significant association between per 10  $\mu\text{g}/\text{m}^3$  increase in  $\text{PM}_{2.5}$  and IHD mortality, with an RR of 1.13 (95% CI: 1.06–1.20) (Fig. 4). Only one cohort study investigated the association of per 10  $\mu\text{g}/\text{m}^3$  increase in  $\text{PM}_{2.5}$  with the IHD incidence. The RR for IHD events was 1.14 (95% CI: 1.08–1.21) when combining the estimated IHD incidence and mortality. No clear association between per 10  $\mu\text{g}/\text{m}^3$  increase in  $\text{PM}_{10}$  and risk of IHD events was observed. The RRs per 10  $\mu\text{g}/\text{m}^3$  increase in  $\text{PM}_{10}$  exposure were 0.99 (95% CI: 0.81–1.18) for IHD incidence, 1.11 (95% CI: 0.96–1.26) for IHD mortality, and 1.08 (95% CI: 0.95, 1.20) for risk of IHD events (Fig C7, Appendix C). Per 10  $\mu\text{g}/\text{m}^3$  increase in  $\text{NO}_2$  exposure was associated with increased risk of IHD events, IHD incidence, and IHD mortality, with RRs of 1.05 (95% CI: 1.04, 1.06), 1.12 (95% CI: 1.02–1.22), 1.05 (95% CI: 1.02–1.07), respectively (Fig C8, Appendix C). There was a weak association between per 9.5 and 11.0 ppb increase in  $\text{O}_3$  exposure and the risk of IHD mortality, with an RR of 1.04 (95% CI: 1.00–1.09) (Fig C9, Appendix C).

The associations between different types of air pollutants and overall and type-specific CVD outcomes were shown in Fig. 5.

### 3.6. Publication bias

We found no significant evidence of publication bias using Egger's linear regression test bias (all  $P > 0.05$ ), except for  $\text{PM}_{2.5}$  with the risk of stroke and CVD ( $P = 0.038$  and  $P = 0.018$ , respectively) (Table B3 in Appendix B). No missing studies were imputed in the funnel plots (Fig D1–D12 in Appendix D).

## 4. Discussion

Overall, our results indicated that per 10  $\mu\text{g}/\text{m}^3$  increase in  $\text{PM}_{2.5}$  exposure could significantly increase 11% risk of CVD events, 12% risk



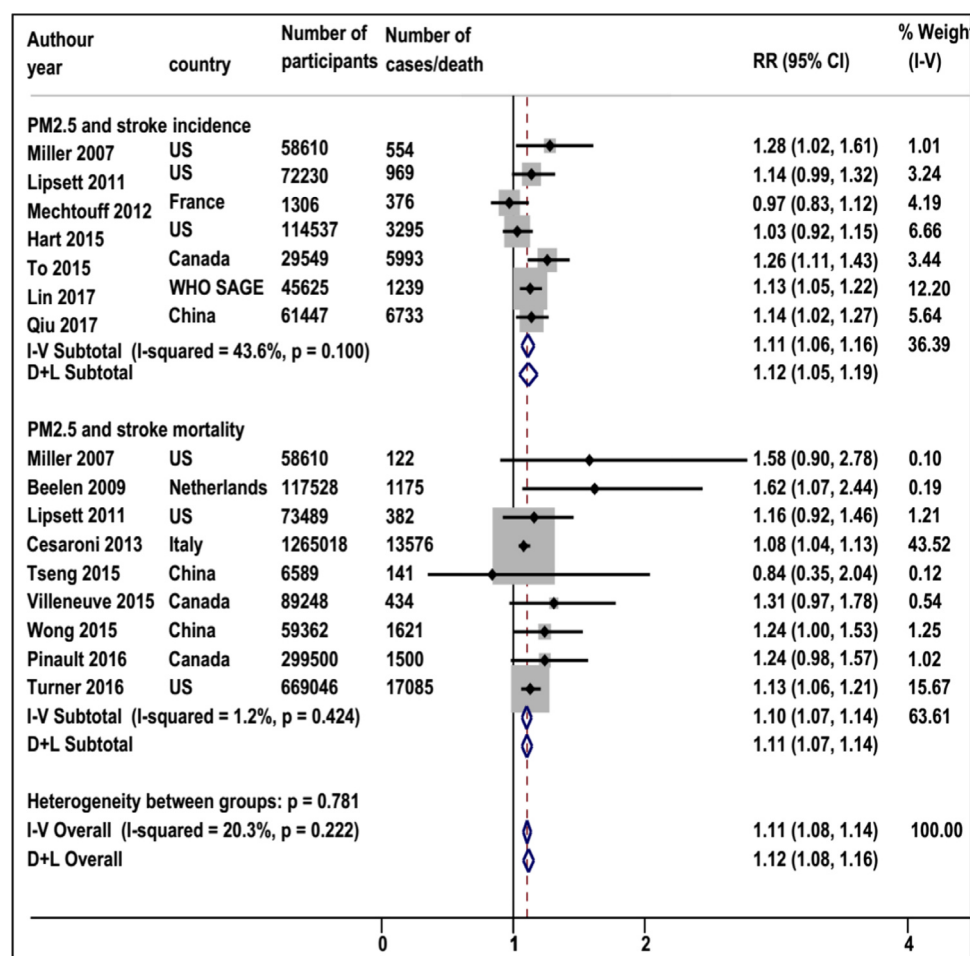


Fig. 3. Relative risk of stroke events (incidence and mortality) per  $10 \mu\text{g}/\text{m}^3$  increase in  $\text{PM}_{2.5}$ . Overall relative risk calculated with both fixed and random effects models.

of stroke events, and 14% risk of IHD events, respectively.  $\text{PM}_{2.5}$ ,  $\text{PM}_{10}$ ,  $\text{NO}_2$ , and  $\text{O}_3$  exposure were separately associated with an increased risk of CVD mortality, and  $\text{NO}_2$  exposure had a greater increase in the risk of CVD mortality. There was a weak association between  $\text{PM}_{10}$  exposure and the risk of CVD events. However, we found there was no clear evidence for the associations between  $\text{PM}_{10}$  exposure and the risk of CVD incidence, stroke incidence, and IHD incidence.

A meta-analysis by Scheers et al. (2015) reported a weak association between per  $10 \mu\text{g}/\text{m}^3$  increase in  $\text{PM}_{10}$  and risk of stroke, with an RR of 1.05 (95% CI: 1.0–1.1). We found that per  $10 \mu\text{g}/\text{m}^3$  increase in  $\text{PM}_{10}$  was not significant with the risk of stroke (RR: 1.04, 95% CI: 0.90–1.17). We observed a stronger association of per  $10 \mu\text{g}/\text{m}^3$  increase in  $\text{PM}_{2.5}$  with the risk of stroke, with an RR of 1.12 (95% CI: 1.02–1.16), compared with the RR of 1.05 (95% CI: 1.01–1.09) Scheers et al. (2015) reported. To focus on the long-term effects of air pollution, we only included cohort studies in our meta-analysis, which might account for the difference between our results and their results. For short-term exposure to  $\text{PM}_{2.5}$  and  $\text{PM}_{10}$ , Shah et al. (2015) found per  $10 \mu\text{g}/\text{m}^3$  increase in  $\text{PM}_{2.5}$  and  $\text{PM}_{10}$  were positively associated with mortality from stroke, with a stronger association for  $\text{PM}_{2.5}$ . We also observed similar results in our meta-analyses. Compared with  $\text{PM}_{10}$ ,  $\text{PM}_{2.5}$  was more susceptible to respiratory uptake, deposition in the lungs, and translocation from the lungs into the circulation (Koulova and Frishman, 2014). It therefore potentially posed a more increased risk of cardiovascular disease.

Faustini et al. (2014) reported the RRs for cardiovascular mortality were 1.13 (95% CI: 1.09–1.18) for per  $10 \mu\text{g}/\text{m}^3$  in  $\text{NO}_2$  and 1.20 (95%

CI: 1.09–1.31) for per  $10 \mu\text{g}/\text{m}^3$  in  $\text{PM}_{2.5}$ . Compared with their analysis, we didn't convert the effect estimates of  $\text{PM}_{10}$  into  $\text{PM}_{2.5}$ ,  $\text{NO}_x$  into  $\text{NO}_2$ , and the different units (ppb and  $\mu\text{g}/\text{m}^3$ ) of  $\text{NO}_2$ , considering the difference of measure and accuracy of effect. While the number of our eligible studies included were less, our results were consistent with theirs. We found that per 9.5–18.4 ppb increase in  $\text{O}_3$  exposure was positively associated with increased risk of stroke mortality and CVD mortality, which was different from the results of Atkinson et al. (2016). The results of CanCHEC cohort study indicated that there was a positive association between per 9.5 ppb increase in  $\text{O}_3$  and CVD mortality, stroke mortality and IHD mortality (Dan et al., 2015). We included the CanCHEC cohort in our meta-analysis, and Atkinson et al. (2016) didn't. It might lead to the inconsistency.

For air pollutant concentrations in the baseline, there was a lack of the standardization and the inconsistency between studies. It might impact the sizes and interpretation of the pooled effect estimates in our analysis, while we selected the specific doses for different air pollutants. Nishiwaki et al. (2013) reported inverse associations between per  $10 \mu\text{g}/\text{m}^3$  increase in PM exposure and the risk of stroke and CVD, respectively. However, their results were interpreted with more caution, when compared with other studies. The air quality standard for the PM in Japan was different from that in other countries, and high salt intake in their study area accounted for a high incidence of stroke. The levels of air pollutants concentration in China was relatively higher than that in the European countries. For the association between air pollutants and CVD mortality, the effect sizes indicated by the studies in China were larger (Cao et al., 2011; Zhang et al., 2014, 2011). However, at

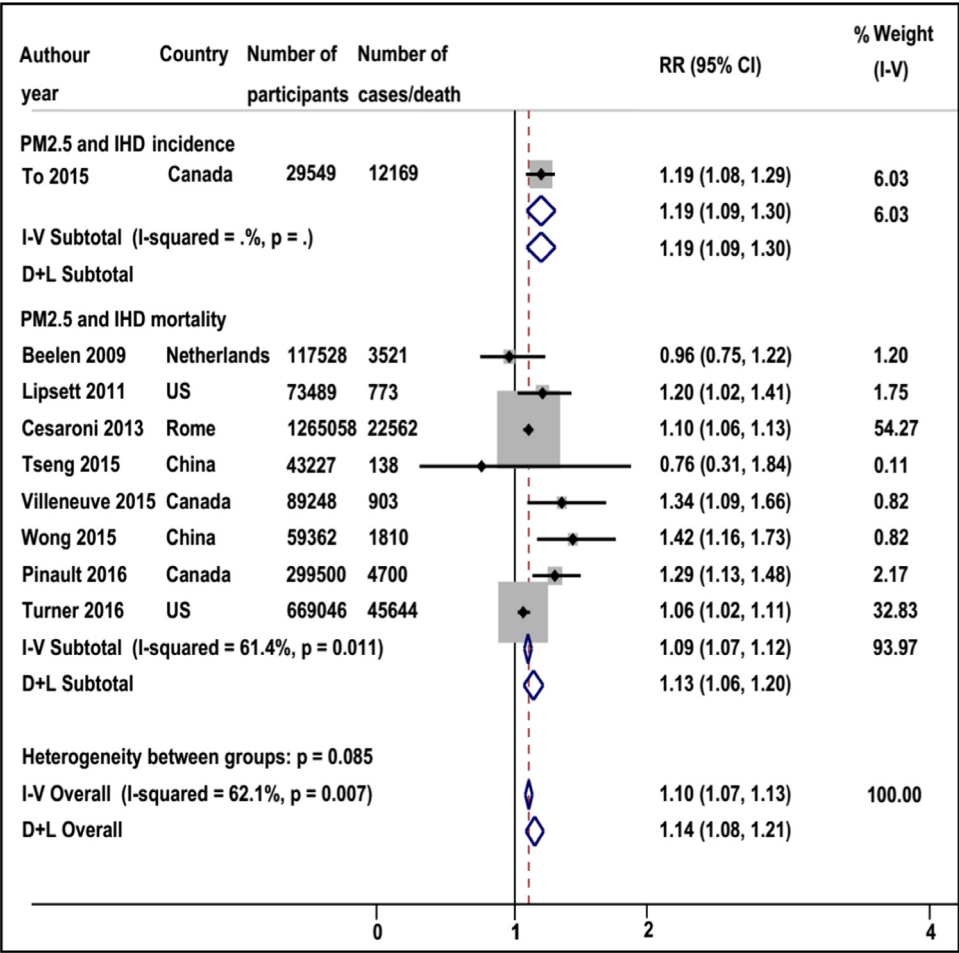


Fig. 4. Relative risk of IHD events (incidence and mortality) per 10 µm³ increase in PM2.5. Overall relative risk calculated with both fixed and random effects models.

present and in this systematic review, the cohorts examining the associations between air pollutants and the risk of CVD were largely different from Europe. The effect estimates in some countries and areas with the high levels of pollution were missing. The different measures of monitoring the air pollutants or the collection of the data across studies might make the effect size not robust. Further and detailed prospective studies would be still needed to assess the association between air pollutants and the risk of CVD.

The associations between smog and disease are complicated in the causes and effects. The association between exposure to air pollutants and the risk of CVD has been demonstrated in both cross-sectional and prospective studies. However, both epidemiological and biological mechanisms that link air pollutants exposures with CVD remain to be further elucidated. Smog can not only cause air pollutants but also other risk factors such as the adverse meteorological factors, lifestyle risk factors, negative psychosocial factors, which may be important factors in the determination of effect estimates of air pollutants. The air pollutants and other health risk factors accompanied by smog may be often interrelated, and may often occur in combination. For example, air pollution exposure is associated with the risk of increased smoking and unhealthy alcohol consumption (Strak et al., 2017). Physical activity in smog days can also amplify respiratory uptake and deposition of air pollutants in the lung (Laeremans et al., 2018). When evaluating the effect of air pollutants on the risk of CVD, the combined and interactional relationship between air pollutants and other risk factors on the risk of health might pose some influence on the effect size for risk of

CVD. The actual effects of the air pollutants, individually and in combination with other risk factors, remain to be disentangled with multidisciplinary research teams in the future.

The strength of the present study was the comprehensive analyses of the associations between air pollutants (PM2.5, PM10, O3, and NO2) and the risk (incidence and mortality) of CVD and CVD subtypes. There were some limitations in our study. The information of health risk thresholds of air pollutants was missing in the included studies, so the effects of the air pollutants on the risk of CVD according to the stratification of pollutant concentrations were not explored in our study. The concentration-response analysis across more exposure doses was not performed because of the limited data. The combined effects of different types of air pollutants on risk of overall and type-specific CVD were not estimated because of lack of relevant well-designed studies.

5. Conclusion

This meta-analysis confirms the evidence that PM2.5 exposure was significantly associated with increased risk of CVD events, stroke events, and IHD events. PM2.5, PM10, NO2, and O3 exposure were separately associated with an increased risk of CVD mortality, and NO2 exposure had a greater increase in the risk of CVD mortality. There was a stronger association between PM2.5 exposure and the risk of stroke and IHD incidence. It urgently needs well-designed studies to further to elaborate the biological and epidemiological mechanisms that link smog with CVD.

|                  | PM <sub>2.5</sub>           | PM <sub>10</sub>            | NO <sub>2</sub>             | O <sub>3</sub>              |
|------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|
| CVD events       | <b>1.11</b><br>(1.07, 1.15) | <b>1.06</b><br>(1.00, 1.12) | <b>1.17</b><br>(1.10, 1.23) | 0.93<br>(0.81, 1.04)        |
| Stroke events    | <b>1.12</b><br>(1.08, 1.16) | 1.04<br>(0.90, 1.17)        | 1.01<br>(0.98, 1.03)        | 0.93<br>(0.76, 1.10)        |
| IHD events       | <b>1.14</b><br>(1.08, 1.21) | 1.08<br>(0.95, 1.20)        | <b>1.05</b><br>(1.04, 1.06) | <b>1.04</b><br>(1.00, 1.09) |
| CVD incidence    | 1.12<br>(0.90, 1.33)        | 0.97<br>(0.65, 1.30)        | 1.01<br>(0.98, 1.04)        | <b>0.63</b><br>(0.58, 0.68) |
| Stroke incidence | <b>1.12</b><br>(1.05, 1.19) | 0.97<br>(0.83, 1.10)        | 1.02<br>(0.94, 1.10)        | 0.81<br>(0.4, 1.22)         |
| IHD incidence    | <b>1.19</b><br>(1.09, 1.30) | 0.99<br>(0.81, 1.18)        | <b>1.12</b><br>(1.02, 1.22) | NA                          |
| CVD mortality    | <b>1.11</b><br>(1.07, 1.14) | 1.09<br>(0.86, 1.29)        | <b>1.23</b><br>(1.15, 1.31) | <b>1.03</b><br>(1.02, 1.05) |
| Stroke mortality | <b>1.11</b><br>(1.07, 1.14) | 1.08<br>(0.86, 1.29)        | 1.01<br>(0.99, 1.03)        | <b>1.02</b><br>(1.00, 1.05) |
| IHD mortality    | <b>1.13</b><br>(1.06, 1.20) | 1.11<br>(0.96, 1.26)        | <b>1.05</b><br>(1.02, 1.07) | <b>1.04</b><br>(1.00, 1.09) |

**Fig. 5.** The summary of associations between air pollutants and CVD (Data in the grids represent the pooled estimates (RR with 95% CI) for the associations by the random effect model. NA represent the effect estimates were not available because of limited data).

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## Author contributions

HXY and YGW contributed to conception and design. All authors were involved in the analysis and interpretation of the data. HXY and SL designed and conducted the statistical analysis. HXY and SL developed the first manuscript draft, which was critically revised for important intellectual content by LS, TW, and YGW. All authors critically reviewed and contributed to the final version, and approved it.

## Conflicts of interest

Declaration of personal interests: none.

## Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.envres.2019.01.040](https://doi.org/10.1016/j.envres.2019.01.040).

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